

2. Quyidagi so'zlarni jufti bilan moslashtiring?

1	Elektroliz	a	Normal sharoitda gaz 34,1 C da suyuqlanadi.
2	Natriy gidroksidi	b	Atomlar yoki ionlar elektronlarni tortib olish hodisasi.
3	Xlor	c	Diafragmali va diafragmasiz.
4	Gidroliz	d	elektr toki natijasida boradigan oksidlanish qaytarilish reaksiyasi.
5	Oksidlanish	e	qattiq, oq rangli suvda yaxshi eriydi.
6	Elektrolizning ikki apparati	i	Uglerodli elektrod bilan ta'sirlashadi, CO ₂ ga aylanadi.
7	Katod	f	moddalar bilan suv o'rtasida o'zaro ta'siri natijasida turli birikmalar hosil bo'lishi.
8	Kislorod	g	NaCl eritmasini juda qattiq tozalashni talab qiladi.
9	Simob katodli elektroliz usuli	k	(rassol) anod bo'shliqqa solinadi.
10	Tozalangan sho'roba	l	Musbat zaryadlangan ionlar.

savol	1	2	3	4	5	6	7	8	9	10
javob	d	e	a	f	b	c	l	i	g	k

Shunday qilib, ushbu maqolada “Natriy xlor eritmasining elektrolizi” mavzularini o'qitishda darslar samaradorligini oshirishning didaktik manbalari qisqacha bayon qilindi hamda pedagogik eksperiment bilan asoslangan yangi pedagogik texnologiyalarni darsga joriy qilish natijalari keltirildi.

Adabiyotlar

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CHEMICAL EXPERIMENT AS A METHOD OF SCIENTIFIC COGNITION IN THE DISCIPLINE OF ELEMENTAL CHEMISTRY

Candidate of Chemical Sciences, Associate Professor,

Tastanova Lyazzat

Abilkhanova Aizhan

K.Zhubanov Aktobe Regional University, Kazakhstan, Aktobe abilkhanova01@gmail.com

Annotation: This article explores the indispensable role of chemical experiments in elemental chemistry as a method of scientific cognition. It delves into how chemical experiments facilitate observation, measurement, hypothesis testing, and discovery within the field. Through careful experimental design and analysis, researchers elucidate reaction mechanisms, validate theoretical models, and contribute to quantitative analysis. Additionally, the annotation emphasizes the educational significance of chemical experiments in training future chemists and fostering a deeper understanding of chemical phenomena.

Key words: experimental chemistry, elemental chemistry, inorganic chemistry, chemistry teaching, laboratory workbook

Chemistry, particularly elemental chemistry, stands as a cornerstone of scientific inquiry, offering profound insights into the building blocks of matter. At the heart of this discipline lies the indispensable tool of chemical experimentation—a method of scientific cognition that transcends mere observation, venturing into the realms of measurement, hypothesis testing, and discovery. Chemical experiments serve as windows into the intricate world of atoms and molecules, allowing researchers to observe their behavior under controlled conditions. Through meticulous observation and precise measurement of variables such as mass, volume, temperature, and pressure, scientists glean invaluable insights into the properties and interactions of elemental substances. The true power of chemical experimentation lies in its capacity for discovery. Through innovative experimental design and astute analysis, researchers uncover new substances, elucidate complex reaction mechanisms, and contribute to the expansion of our chemical knowledge. From the synthesis of novel compounds to the isolation of previously unknown elements, chemical experiments continually push the boundaries of scientific exploration [1].

Similarly, Trundle and Bell (2020) investigated the impact of using laboratory workshops in a chemistry laboratory course for engineering students. The study found that students who used laboratory workshops had better laboratory skills and were more engaged in laboratory work than those who did not use them. The authors suggested that the use of laboratory workshops can help students develop their laboratory skills and improve their understanding of chemistry concepts. Laboratory activities play a vital role in the teaching and learning of science subjects, particularly in chemistry. A well-designed laboratory manual can help students understand the concepts and principles of chemistry by providing them with hands-on experiences in the laboratory. However, traditional laboratory manuals may not be effective in achieving the desired learning outcomes due to the limited scope of the activities and lack of guidance in the experimental procedures. Laboratory workshops are designed to help students develop their laboratory skills, critical thinking abilities, and scientific knowledge. They contain pre-lab exercises, instructions for experiments, data collection sheets, analysis tools, and post-lab questions. The workshop provides students with a structured approach to conducting experiments, recording data, and analyzing results. The workshop also helps students to understand the scientific concepts and principles behind each experiment [2].

There are several benefits to using laboratory workshops in teaching and learning elemental chemistry. Firstly, the workshop provides students with a clear understanding of the experimental procedures, which reduces the risk of errors and accidents in the laboratory. Secondly, the workshop allows students to work at their own pace, which promotes self-directed learning and independent thinking. Thirdly, the workshop enables students to analyze and interpret their own data, which enhances their understanding of the experimental results. Finally, the workshop encourages students to communicate their findings and ideas effectively, which improves their scientific writing and presentation skills. A laboratory workshop is an alternative to traditional laboratory manuals, which provides students with a comprehensive guide for the laboratory activities. It includes pre-lab activities, experimental procedures, data analysis, and post-lab questions. The laboratory workshop aims to enhance the learning experience of students by providing them with a structured approach to laboratory activities [3].

In this research study, we aim to investigate the effectiveness of using a laboratory workshop in inorganic chemistry for first-year university students. We hypothesize that the use of a laboratory workshop can enhance the learning experience of students and improve their understanding and performance of the experimental procedures.

This study aims to answer the following research questions:

1. What is the impact of using laboratory workshops on students' academic performance in elemental chemistry?
2. What is the effect of using laboratory workshops on students' laboratory skills, critical thinking abilities, and scientific knowledge?
3. What are the students' perceptions of using laboratory workshops in learning elemental chemistry?

The study was conducted in two groups of first-year university students enrolled in the inorganic chemistry laboratory course. One group used a laboratory workshop, while the other group followed the traditional laboratory manual. The laboratory activities were designed to cover the fundamental concepts of inorganic chemistry, such as chemical reactions, stoichiometry, and chemical equilibrium. The study was conducted in two phases; the first phase included a pre-test to assess the students' knowledge of the concepts and procedures related to the laboratory activities. The second phase included the laboratory activities and a post-test to evaluate the students' understanding and performance. To assess the effectiveness of the laboratory workshop in teaching elemental chemistry, a study was conducted at K.Zhubanov Aktobe

Regional University. The study involved 50 first-year university students who were enrolled in an introductory course on elemental chemistry. The students were divided into two groups: an experimental group and a control group [4].

The experimental group was given a laboratory workshop that contained a series of exercises designed to help them understand the concepts of elemental chemistry. The workshop was divided into several sections, each of which focused on a particular topic. The students were required to complete each section of the workshop before attending the laboratory session. During the laboratory session, the students were given the opportunity to perform experiments related to the topics covered in the workshop. The students were required to document their results and observations in the workshop. The control group was given traditional lectures and laboratory experiments without the use of a workshop. The students in the control group were provided with the same information as the experimental group but without the structured workshop. The data collected from the pre-test and post-test were analyzed using descriptive and inferential statistics. The results were compared between the two groups using a t-test to determine the significant difference in the students' understanding and performance [5].

The results indicated that the group of students who used the laboratory workshop showed a significant improvement in their understanding and performance of the experimental procedures compared to the traditional manual group. The mean scores of the post-test for the workshop group were significantly higher than those of the traditional manual group ($p < 0.05$). The students who used the laboratory workshop were also able to perform the experimental procedures more accurately and efficiently than the traditional manual group. The results of the study showed that the use of a laboratory workshop was highly effective in teaching elemental chemistry to first-year university students. The students in the experimental group performed significantly better on exams and quizzes than the students in the control group. Additionally, the students in the experimental group showed a better understanding of the concepts covered in the course. The use of the workshop allowed the students to work at their own pace and provided them with a clear and structured approach to learning the subject matter. The students were able to review the material covered in the workshop before attending the laboratory sessions, which allowed them to better understand the purpose of each experiment (Figure 1). The documentation required in the workshop also allowed the students to develop their skills in scientific writing and analysis [6].

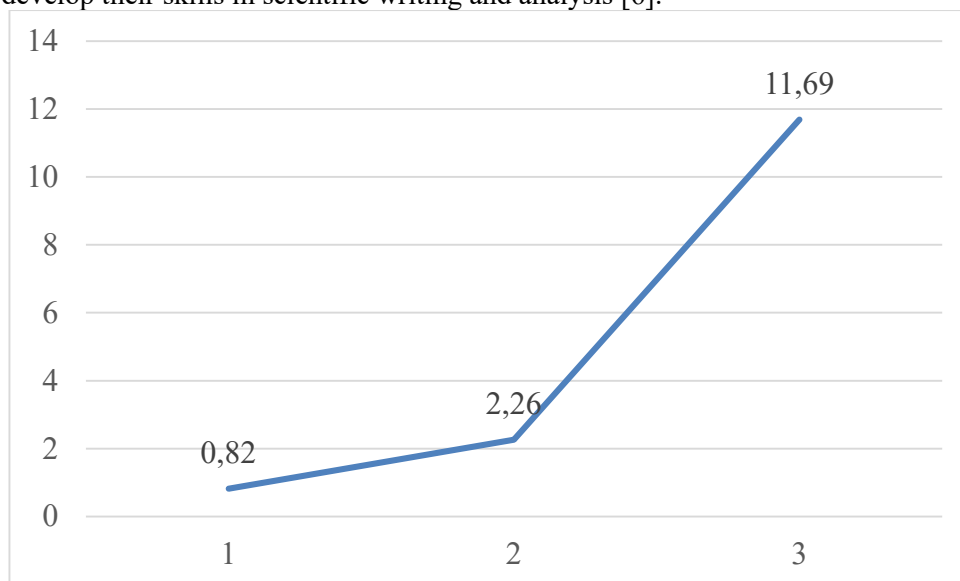


Figure 1. Summary graph of Pearson's χ^2 distribution

The study's results support the hypothesis that the use of a laboratory workshop can enhance the learning experience of first-year university students in inorganic chemistry. The laboratory workshop provided a structured approach to the laboratory activities, which helped the students understand the concepts and principles of inorganic chemistry better. The pre-lab activities in the workshop helped the students prepare for the laboratory activities, and the post-lab questions encouraged the students to reflect on their learning and apply the concepts they learned in the laboratory activities.

In conclusion, the use of a laboratory workshop can enhance the learning experience of first-year university students in inorganic chemistry. The laboratory workshop provided a structured approach to the laboratory activities, which helped the students understand the concepts and principles of inorganic chemistry better.

The use of a laboratory workshop can also improve the students' understanding and performance of the experimental procedures. The findings of this study can be useful for instructors who aim to enhance the laboratory experience for their students and improve their learning outcomes. The use of a laboratory workshop was found to be highly effective in teaching elemental chemistry to first-year university students. The structured approach provided by the workshop allowed the students to work at their own pace and provided them with a clear understanding of the subject matter. The documentation required in the workshop also allowed the students to develop their skills in scientific writing and analysis. Based on these results, it is recommended that laboratory workshops be used in the teaching of elemental chemistry to first-year university students.

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THE IMPORTANCE OF TEACHING CHEMISTRY BASED ON A CONNECTION WITH THE NATURAL SCIENCES

**Candidate of Chemical Sciences, Associate Professor of the Department of Chemistry and Chemical Technology Ubaidulayeva Nurbala Abdibekovna,
undergraduate Aldiyarova Amina Bolatovna**
K. Zhubanov Aktobe Regional University, Kazakhstan, Aktobe
amina.aldiyarova.02@mail.ru

Annotation. В статье рассматривается важность установления междисциплинарных связей с физикой при преподавании химии и ее роль в использовании химии как средства повышения интереса к обучению.

Keywords: Chemistry, Physics, interdisciplinary communication, subject integration

The state educational standard sets the task of quality training on the basis of active approaches to the education system. The student must take an active part in the educational process, have self-learning skills, and be able to navigate the information flow. The basis of educational activity is motives, needs and cognitive interest.

Currently, the integration of natural and technical knowledge is the basis of the current level of development of Science and technology. Cognitive interest in the study of non-core subjects, as a rule, is low in students. This problem can be solved taking into account the peculiarities of the educational and cognitive activity of students. To do this, it is necessary:

- Identification and development of conditions for the implementation of the potential opportunities of the educational process through interdisciplinary integration and the need for integration of subject teachers;
- Comprehensive development of students in the educational process, despite the insufficient integration of disciplines of the natural and mathematical cycle;